

pubmed_25433282 ? injury

Remote renal ischemia-reperfusion injury (IRI) following infra-renal aortic occlusion leads to acute kidney injury and systemic inflammation. Hydrogen sulfide is a mediator of IRI and can ameliorate tissue injury in many organ systems. Its role in vascular surgery has yet to be established. We assessed the role of hydrogen sulfide in a rodent model of aortic occlusion. Wistar rats were divided into sham, control, and treatment groups (n = 6). Inflammation was assessed using a nonrecovery protocol. The infra-renal aorta was cross-clamped for 60 min and animals were reperfused for 120 min. Ten minutes before clamp release, treatment animals received hydrogen sulfide (10, 30, or 50 ?g/kg) and control animals received 0.9% saline injected into the retroperitoneum. Renal injury and histology were assessed by a recovery protocol. The procedure was identical to the nonrecovery arm but with a single dose of hydrogen sulfide (30 ?g/kg) and animals were recovered for 7 days. There was no difference in animal weight between the groups (P = 0.337). In the nonrecovery arm, there was a reduction in serum levels of tumor necrosis factor alpha in sulfide-treated animals compared with controls (909 ± 98 vs. 607 ± 159 pg/mL; P = 0.0038). There was also a reduction in myeloperoxidase-positive cells in renal tissue in the sulfide-treated animals compared with controls (8 ± 4 vs. 17 ± 9 ; P = 0.03). There was no difference in histological injury score or endothelin-1 levels. In the recovery arm, there was no difference in renal function, Kidney Injury Molecule-1 levels, or histological injury scores.

Question: Does hydrogen sulfide reduce inflammation following abdominal aortic occlusion in rats?

Answer: Hydrogen sulfide has systemic and renal anti-inflammatory effects in remote IRI following aortic occlusion in rats.